

New Construction Report Card, Part II

Complex building designs make even the best-laid plans for home performance hard to fulfill.

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Building a house that is energy efficient, healthy, and comfortable is quite challenging (otherwise, *Home Energy* wouldn't exist). House designs and building technologies are growing in complexity, increasing the difficulty of successfully integrating all the components into a workable system. A California Energy Commission (CEC)-sponsored project completed in 2002 involved detailed diagnostic testing of building envelopes and HVAC system performance to assess how key building components are being integrated in 60 new California production homes.

Results of the testing from the Residential Construction Quality Assessment (RCQA) project are valuable in quantifying the performance of these homes. They also provide information on various testing procedures and the assumptions used in the California building standards. Diagnostic techniques and visual inspections were used to assess duct leakage, HVAC system air flow, quality of insulation installation, envelope airtightness, and other building features that affect energy use and comfort.

A first set of 30 homes tested was selected to assess California leading-edge practitioners who are installing, testing, and verifying tight-duct distribution systems. The results of these tests, which were held from September 1999 to January 2000, were presented in the last issue of *Home Energy* (see "New Construction Report Card," Jan/Feb '03, p. 18).

Thirty additional homes were tested from November 2001 to February 2002. One of the key site selection criteria was to test homes that had taken the tight-duct credit available under the 1998 Title-24 residential building standards. The 1998 standards provide a credit for instal-



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This is a common draftstopping problem—an interior wall cavity open to the attic.

lation and verification of duct systems with leakage not exceeding 6% of the default system fan flow (nominal 400 CFM per ton). The duct credit requires that the installing HVAC contractor and a third-party HERS rater test the duct system and document leakage at less than the 6% level. All the houses tested were involved in city, utility, large-scale, or contractor-sponsored energy efficiency programs requiring some level of diagnostic testing (see "The Cream of the Crop").

Testing Protocol

As in any field research and testing project, several attempts to assess the first 30 homes were found to be less useful than anticipated. Lessons learned in the first round of testing resulted in the following changes to our testing protocol.

Use of alternative duct leakage diagnostic tools. Three alternative duct leakage tests were originally used in addition to the common duct pressurization test (Duct Blaster at 25 Pa). The pressure

pan test, the blower door subtraction method, and the house pressure test were all found to provide inconsistent assessments of whether or not the duct system was tight. Our conclusion was that although these tests might be useful for typical leaky duct systems, they did not provide a reliable assessment for the tight-duct houses we were testing.

For the second round of testing, Lawrence Berkeley National Laboratory's DeltaQ test was performed at several of the test sites. DeltaQ is an alternative method for calculating both supply and return duct leakage under normal system operating conditions (see "An Easier Way to Measure Duct Leaks," *HE* Sept/Oct '02, p. 34). The DeltaQ procedure uses a blower door to measure flow differences at various house pressures with the air handler on and off. Since the DeltaQ procedure calculates duct leakage during air handler operation, it should provide a more accurate measurement of real duct leakage than the standard 25 Pa duct pressurization test.

The Cream of the Crop

The houses tested were built with the help of utility programs, third-party programs, or individual contractors, with tight ducts in mind (see Table A for program summaries). All of the houses fell into one of the following categories:

- Projects claiming the Title 24 tight-duct credit. These houses were identified primarily through various HERS testing organizations and major Title

24 compliance consultants.

- Builders participating in utility programs that require verified tight ducts (PG&E's Comfort Home program, Sacramento Municipal Utility District's Advantage Home program, Southern California Gas Company's Energy Advantage Home program, and Edison's Comfortwise program).
- Builders participating in a third-party quality construction program that

includes diagnostic testing and guaranteed energy bills (MASCO's EFL program).

- Builders participating in a municipal program that encourages energy-efficient construction practices and requires duct testing or sampling (City of Irvine).

- An exemplary "independent" house with guaranteed energy bills.

Table A. Summary of Duct Leakage Requirements by Program Type

Program	Leakage Target	Verification Component
Alternative calculation method	6% of 400 CFM/ton AC capacity or flow hood-measured fan flow	3rd party testing on all houses
PG&E Comfort	Home 6% of nominal fan flow	Almost exclusively contractor testing (some 3rd party); ~20% of floor plans ultimately retested by PG&E
City of Irvine	6% of nominal fan flow	3rd party sampling of houses with sampling rate based on recent rate of compliance with target
SCG Energy Advantage Home	6% of nominal fan flow	100% contractor testing
SMUD Advantage Home	6% of nominal fan flow	100% contractor testing; spot in-house observation of contractor testing
Environments for Living	5% of nominal fan flow	3rd party testing on 1 in 7 houses
ComfortWise	6% of nominal fan flow	3rd party testing on roughly 1 in 4 houses
Builder	A "tight" duct system	Contractor testing

The Energy Conservatory's TrueFlow air handler flowmeter was added as a second method for measuring HVAC system air flow. The TrueFlow device (see "Flow Control Gains in Accuracy," *HE* Mar/Apr '02, p. 34) is easily installed in the filter slot at the return air grille and provided a comparison for the powered flow hood technique used throughout the project. (The powered flow hood technique requires measuring supply plenum static pressure during HVAC system operation. The Duct Blaster is then connected to the system return and the Duct Blaster fan is adjusted until the supply plenum pressure matches the prior static pressure reading. Air flow through the Duct Blaster is then recorded.)

Quantifying wall and ceiling insulation defects. In the first round of testing, the infrared camera did not provide satisfactory quantification of

wall cavity R-value, due to difficulties in consistently achieving adequate indoor-to-outdoor temperature differences. For the second round of testing, the CEC was interested in evaluating another approach. The contract team developed a visual inspection procedure for quantifying wall thermal performance by accounting for insulation voids and batt compression. Real wall performance is degraded by two factors: increased wall cavity framing and insulation installation defects, including compressed insulation due to wiring and plumbing, shoddy installation, voids, rounded shoulders at exterior cavity corners, and insulation stuffed into narrow cavities.

The visual inspection procedure utilized a wall insulation takeoff form to quantify the R-value degradation as a function of the degree of compression. The inspector begins at the front door of an insulated (but not yet drywalled)

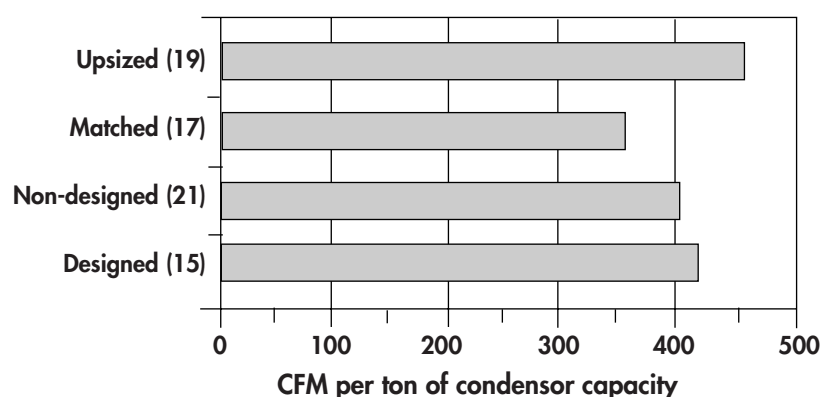
house, and works around the perimeter of the house. For each new exterior wall section, the inspector notes

- nominal wall construction (for example, 2 x 4, 16 inch on center, R-13 cavity, R-4 exterior);
- gross wall dimensions (length x height);
- area of any windows or doors; and
- defect characterization (area and R-value).

Multiple defects can be entered for each wall section. Void areas are calculated at zero-cavity R-value and optimally performing sections are recorded at nominal cavity R-value. Other insulation defects require the user to assign a cavity R-value (based on average percent compression) for the defect and an associated defect area. Each wall section is represented by a summation of subareas, as shown in Equation 1, which total the net wall area for that section.

Table 1. Installed Cooling System Sizing

Location	Number of houses	Mean Sizing (ft ² /ton)	Minimum (ft ² /ton)	Maximum (ft ² /ton)
El Toro	2	489	477	500
Burbank	1	550	n/a	n/a
Red Bluff	7	601	448	1003
Sacramento	14	618	475	820
Fresno	2	401	400	401
Palm Springs	4	431	361	486
Average	30	543	361	1003

Measured HVAC Airflow

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Figure 1. Upsizing the air handler and detailed mechanical design both increase air flow.

Equation 1: Wall Section UA =

$$\sum_{i=1}^n U_i A_i + U_2 A_2 + \dots + U_i A_i$$

In Equation 1, U_i is the parallel path calculated U-value based on cavity R-value and wall construction characteristics and A_i is the defect area. The cavity R-value can be calculated from the whole-house wall UA since the cavity R-value is the only unknown in the parallel path U-value calculation methodology.

Detailed evaluation of duct system. A visual inspection was added to assess whether R-8 ducts could be used at each house, and to determine whether any duct runs were too long. Excess duct length was calculated by assessing whether duct runs could be shortened (by moving registers from exterior walls to interior walls, by replacing multiple registers with one register, or by eliminating registers entirely in

small areas such as halls or open laundry rooms). In addition, the contract team completed a detailed *Manual J* and *D* analysis to see whether duct and cooling equipment sizing in each house conformed to the standard.

Evaluation of fireplace leakage.

In the first round of testing, a blower door test was completed with the fireplace sealed at only one of the 30 sites. The high measured leakage attributed to the fireplace led us to study this factor in greater detail. All the houses in the second round of testing were tested for fireplace leakage by performing blower door measurements before and after sealing the fireplace doors.

A Snapshot of the 30 Houses

Twenty-three of the test houses were located in the California's Central Valley, where a significant portion of new home construction is occurring. They ranged from Redding in the far north of the val-

ley to Bakersfield in the far south. The remaining 7 houses were located in Southern California, ranging from central Los Angeles eastward to Palm Desert. The floor area of the 30 houses ranged from 1,120 ft² to 4,170 ft²; the mean floor area was 2,135 ft². All but 2 houses were slab-on-grade construction. There were 22 single-story and 8 two-story houses.

Twenty-seven of the 30 houses were 2 x 4 framed wall construction, with studs nominally spaced on 16-inch centers. Cavity R-values ranged from R-13 to R-15, and 16 of the 27 houses had rigid exterior insulation. The 3 remaining houses were 2 x 6 construction with studs on 16-inch centers and cavity R-values of R-20 to R-21. Three of the houses had spray-applied cellulose in the wall cavities and 1 had blown-in fiberglass. Twenty-four of the houses had loose-fill blown-in ceiling insulation (except in vaulted ceiling areas, where batts were often installed). In 20 of these houses fiberglass insulation was used. In the other 4, ceiling insulation was cellulose. Four of the houses were Platinum rated by MASCO's Environments for Living (EFL) program with R-22 blown-in insulation at the roof rafters (see "La Crème de la Crème," p. 39). Ceiling R-values for the 30 houses were generally R-30 and R-38, with a range of R-19 to R-49.

All the houses tested had split-system air conditioning units with gas furnaces. Four of the houses had 2 HVAC units and 1 had 3, resulting in a total of 36 HVAC units. Twenty-seven of the HVAC units were installed in unconditioned attics, 4 were installed in conditioned (cathedralized) attics, 3 in indoor closets, 1 in an outdoor closet, and 1 in a garage. More than half (20 of 36) of the cooling systems had thermal expansion valves (TXVs), as compared to only one-third of the cooling systems in the first 30 houses tested. TXV-equipped cooling systems have better refrigerant metering ability and can better maintain cooling system performance if the refrigerant charge deviates from the manufacturer's specifications.

HVAC Performance

The inspection and testing of the 30 houses revealed a number of interesting findings.

Ducts. One of our tasks was to determine whether there was enough room to install R-8 ducts instead of the standard R-4.2 ducts. For interior ducts running through duct chases, it was often not possible to tell whether R-8 ducts could fit within existing cavities. However, in all cases where visible inspection was possible, it was determined that R-8 duct could be accommodated.

At 23 of the 30 sites, the supply boots were taped at the register to prevent leakage at the interface between the supply boot and the adjacent drywall. At 1 of the 7 remaining sites, the register boots were not taped. At the other 6 sites, verification was not possible without damaging the paint seal between the ceiling and the grille.

Visual inspection revealed that, in the majority of houses, installed duct length could be reduced (by shortening duct runs, by eliminating redundant ducts to a large room, or by eliminating an entire run in some cases). Of the 21 houses where duct length could be reduced, the average percentage of length reduction was 27%, indicating a fairly common problem of installing too large a duct system. Reducing duct length reduces duct conductive heat transfer to the attic and reduces duct friction losses, increasing total supply air flow. In one case, the potential duct length reduction would reduce duct conductive losses by over 1 ton at design cooling conditions.

Duct systems were visually inspected for installation defects, R-value, and length and diameter by duct section (for area takeoffs). We found very few duct installation problems related to lack of duct support, kinks and bends, compression between truss members, and restrictions caused by supports or framing members. In a few cases, we observed restriction at duct support points where the bend was too sharp. No leakage points were identified during the duct visual inspection. When we tested the first 30 homes, we also found a very low frequency of duct installation problems.

Cooling system sizing. Sizing of the cooling systems averaged 543 ft² per ton; it ranged from 361 ft² per ton in the desert near Palm Springs to 1,003 ft² per ton in Red Bluff, near Redding (see Table 1). Two cathedralized attic houses in Tracy were sized at over 750 ft² per

ton. Removing these three outliers from the sample reduces the average system sizing to 515 ft² per ton. *Manual J* sizings completed by our project team engineers indicated an average cooling system oversizing of 10%. Systems designed by the mechanical contractor (defined as sites where mechanical design was included as part of the plan submittal) were only 2% oversized, while



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This subpar fiberglass batt insulation job shows three problems that affect performance: the insulation is not fully lofted so that it will touch the drywall air-barrier once it is installed, it is compressed behind the wiring and in narrow cavities, and a void is created in front of the wiring.

nondesign systems were 15% oversized. Since *Manual J* assumes substantially tight ducts, the real world (approximately 15%) design load variation between “industry standard” and “tight ducts” would increase average oversizing to 29%. Based on this sample of houses, it is clear that the use of *Manual J* leads to better HVAC system sizing, although further improvements are needed in the *Manual J* methodology to properly credit tight duct systems and eliminate conservative assumptions elsewhere in the methodology.

Air flow. As in our first round of tests, we found that there was a general problem in getting air to where it was supposed to be. *Manual D* duct sizing and air flow measurements clearly demonstrate consistent undersizing of both return ducts and larger-diameter supply ducts. Improved duct sizing combined with shorter duct systems would reduce overall duct friction and fan energy, and increase system air flow. HVAC system air flow for these 30 houses averaged 412 CFM per ton, or 3% more than the industry standard of 400 CFM per ton, and an 18% increase

in CFM per ton over the first 30 houses tested. This was largely because it has become more common to upsize the indoor air-handling unit (fan, furnace, and coils) while keeping the capacity of the outdoor condensing unit the same, or in some cases reducing its size. This practice has the advantage of providing greater air flow and sensible cooling capacity, although air handler fan power is increased (see Figure 1). As one might expect, both upsizing the air handler and detailed mechanical design were shown to increase HVAC air flow. The 19 upsized units provided an average 457 CFM per ton of condenser capacity, compared to 362 CFM per ton for matched systems.

Duct leakage. Average measured duct leakage was found to be 6.3% of default system fan flow, slightly higher than the Energy Commission’s tight-duct target of 6%. Virtually all of the systems had been previously tested. Eighteen of the 36 systems met the Energy Commission’s 6% target and another 4 systems were under 7%. Duct systems that had been previously tested either by the installing HVAC contractor or by a third-party tester were found by our testing team to have 5 CFM higher average duct leakage than was measured by the previous testing. Mastic was found on more systems in this second set of 30 houses than were found in the first set, and only in one house had cloth-backed duct tape been used. The duct sealing technique incorporating drawbands and mastic was found to be the tightest of the sealing systems.

The DeltaQ test was found to provide generally good agreement with our technique for disaggregating duct leakage. The DeltaQ test is probably more accurate than ours, but it requires the use of a blower door, which is not as common as the Duct Blaster.

Envelope leakage. Blower door leakage tests for the 30 houses revealed an average specific leakage area (SLA; leakage normalized by floor area) of 2.96 square inches/ft². Ten of the 30 houses were removed from the sample because they had house wrap installed, or because they had participated in MASCO’s EFL program, which places an emphasis on reducing envelope leakage. The SLAs from the remaining 20

houses were normalized by adjusting the measured envelope leakage to represent a typical house (based on project results of one wood-burning fireplace and an average of 12 recessed lights, each leaking an average 8.7 CFM₅₀). The average adjusted SLA for the 20 houses was 3.22; SLAs ranged from 1.99 to 3.93. The 3.22 SLA is 8% lower than the average for the first 30 houses that we tested. The considerable range in SLA (standard deviation of 0.45), even after adjusting for fireplace and recessed lights, reflects the random nature of house leakage, which is due to generally poor draftstopping on vertical chases, interior columns, and drop ceilings. Pressure readings of interior building cavities during 50 Pa depressurization can help to determine the presence of draftstopping problems, but they cannot quantify the magnitude of the air flow through these air barrier defects. The variability in the magnitude of these defects is evidenced by the sizable difference in SLA (>12%) between five identical house pairs described in the earlier article, cited above.

Fireplace leakage measurements were completed for all houses in the second round of testing. Wood-burning fireplaces were found in 12 houses and leaked an average of 173 CFM₅₀. Gas burning fireplaces, found in 7 houses, leaked an average of 64 CFM₅₀. Direct vent gas fireplaces with sealed combustion air and tightly fitting doors were found in 9 houses, and leaked an average of only 31 CFM₅₀.

Insulation installation effectiveness. Ten houses were analyzed using the wall insulation inspection methodology. Of the ten, five



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(top) This is an ideal wall insulation. The cavity is filled and there are no voids. (bottom) This is the way attic insulation should look—it has the proper thickness and is nice and even.

represented industry standard workmanship and five represented high-quality workmanship. For the five industry standard sites, the observed installation defects degraded the in situ cavity R-value to an average of 69% of nominal. The five high-quality sites performed

much better, with an average cavity R-value that was 94% of nominal.

Blown-in ceiling insulation was found, on average, to be 99% of the required depth. Variations among sites were substantial, with an average standard deviation of 19%. To achieve performance consistent with design intent, greater attention must be paid to enforcement to ensure adequate blown-in ceiling insulation depth and proper density in all new homes.

The first part of our study showed that the envelope protocols developed for the CEC are useful in documenting the framing and air barrier installation procedures that contractors should be following. However, there is often a disconnect between the envelope protocols and what is happening in the field. Generally, installers are not taking sufficient care in installing insulation. For example, they are not slitting batts around wiring and plumbing, or properly supporting and air sealing kneewall and skylight shaft insulation. Airtight draft stops were seldom installed properly, with the result that interior (uninsulated) wall cavities were thermally well connected to the attic. This problem is especially prevalent in houses with complicated architectural features, such as arches, columns, frequent transitions in ceiling height, and cantilevered floors.

The Best Could Be Better

The increasing architectural complexity of new homes requires greater

La Crème de la Crème

MASCO's EFL program provides homeowner energy bill guarantees, contractor training, and improved construction quality procedures, including diagnostic testing. We found that the program, which is fairly new to California, was delivering a higher-quality product to the marketplace by emphasizing construction details that are frequently overlooked. One such detail was the cathedralized attic. This concept, developed by Building Science Corporation, involves creating a

semiconditioned attic space by sealing the attic and insulating at the underside of the roof. By locating ducts in conditioned space, duct leakage and conduction losses are largely eliminated.

EFL contractors undergo training performed by Advanced Energy Corporation. This training addresses insulation installation, air barrier details, pressure balancing, fresh-air ventilation, indoor air quality, duct leakage, and diagnostic testing. Field supervision of subcontractors exceeds that on typical production housing pro-

jects. More extensive diagnostic testing (duct and envelope leakage and interior pressure differences during HVAC operation) provides additional feedback to the builder on the quality of their homes. The EFL program provides three levels of house performance: silver, gold, and platinum. The platinum level is designated as a DOE Building America home. Four of the homes tested in the RCQA project were gold level and four were platinum level (for more information, go to www.eflhome.com).

vigilance on the part of framers, insulators, and drywall contractors to create a single thermal/pressure boundary between conditioned and unconditioned space. The more complex the design, the more coordination is needed among the various members of the design team. Yet, mechanical contractors are rarely consulted regarding the integration of ducts and HVAC equipment into the house design. Contractors often lack both the knowledge and the time to implement “house as a system” construction concepts. In addition, there is not an adequate infrastructure in place to provide contractors and installers with the necessary training and certification.

Signs of change are starting to appear. One builder whose homes were tested received two insulation bids for an upcoming construction phase in a subdivision. One bid was for industry standard work. The other bid (roughly 30% higher) was for installation consistent with EFL’s zero defect insulation standard. Once builders become aware of the benefits, they may be willing to pay a bit more for improved insulation installation and attention to air barrier details to avoid call-back and potential litigation problems down the road. As the mechanical contractor becomes comfortable with the improved quality of building envelopes, HVAC cooling system oversizing should be reduced, saving the builder money. Enterprising builders are becoming aware of these developments, and are seeing construction quality as a way to differentiate themselves from their competitors.

The testing of the 60 new California homes in the RCQA project offered a glimpse of both the good and the bad in residential construction. We found—on the positive side—that the HVAC industry is getting better at delivering tight duct systems. The increasing use of mastic and mechanical fasteners, the advent of programs such as EFL that require contractor training and increased diagnostic testing, and more and more builders and construction superintendents who care about energy efficiency have all contributed to this improvement.

We found—on the negative side—that increasing architectural complexity frequently results in draftstopping prob-

lems that degrade envelope performance by providing a strong thermal connection between conditioned and unconditioned space.

The results of the RCQA project indicate that significant improvements in cavity insulation performance can be achieved. Duct design and layout can be improved to eliminate duct systems that are oversized in length but undersized in diameter, resulting in improved air flow and room-by-room air distribution. Improved HVAC sizing tools and increased use of *Manuals J and D* by mechanical contractors will also improve home performance. Improving the other building elements investigated in the RCQA project (and others that were not addressed) will contribute to improved performance, comfort, and indoor air quality for California production housing in the years ahead.

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For more information:

For more information on the RCQA project, contact Bruce Maeda of the California Energy Commission. Bruce’s phone number is (916)654-4077 and his e-mail is bmaeda@energy.state.ca.us.

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